

Problem Set 2

Separable Differential Equations

PSet Tips

- ◆ Remember to explain the reasoning behind all of your answers.
- ◆ Keep your work organized and assign a page to each problem on Gradescope.
- ◆ You don't have to do this by yourself — office hours, the MQC, and your fellow students are great resources if you get stuck.

In problems 1–4, find the general solution to the differential equations.

1 $\frac{dy}{dx} = \frac{x^2}{y^2}$

2 $\frac{dy}{dx} = 3y$

3 $\frac{dy}{dx} - \sqrt{4xy} = 0$

4 $\frac{dy}{dx} - (1+x)(1+y) = 0.$

In problems 5–8, find the particular solution to the differential equation.

5 $\frac{dy}{dx} = \frac{2x + 3x^2}{3y^2}; \quad y(1) = 2.$

6 $\frac{dy}{dx} - 2e^{(x+2y)} = 0; \quad y(0) = 1.$

7 $\frac{dy}{dx} + 3x^2y^2 + 2xy^2 = 0; \quad y(1) = 3.$

8 $\frac{dy}{dx} = 3x^2(y^2 + 1); \quad y(0) = 1.$

9 A population of bacteria grows at a rate proportional to the size of the existing population. Suppose that the population grows to 10 times its original size in 24 hours. How long did it take for the population to double?

10 Suppose you invest \$10,000 in an account that grows at a rate of 4% per year compounded continuously.

- (a) How long does it take for the investment to double its original amount?
- (b) How much money will be in the account after 10 years?

11 A turkey is heated in an oven to 200°F . At time $t = 0$ it is taken out of the oven and put on a counter in a room whose temperature is 70°F . After 20 minutes the turkey's temperature is 150°F . At what time will the turkey's temperature be 125°F ?